ENG499 – Sensor Data Quality Report (CSE-1)

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1. What I did

I analysed the 25 temperature loggers in *sensor_readings.csv.gz* (2 in the cold room, 3 on vehicles, 20 in store cabinets, one reading every 5 minutes), using *sensors.csv* for the sensor types. I looked at 1 April to 30 September 2025, which is 183 days or 52,704 expected readings per sensor. The 7 warm-up days in March are left out. To find which vehicle readings are real transport time, I also used the event log and the shipment units. All steps are in one Python script, **sensor_cleaning.py**, which I can run again on any new data. It saves a summary table, figures and a cleaned file.

2. Missing readings and gaps

For each sensor I compared the readings I have with the 5-minute slots I expect. A gap is a run of slots with no reading. The fixed sensors lose between 0.73 % and 0.90 % of their readings. TS-U12 is higher (1.77 %) because of one long outage that I describe in section 4. The vehicle loggers lose about 3 % (V1 2.95 %, V2 2.91 %) and V3 loses 4.26 %. V3 has 124 gaps of 15 minutes or longer and the longest is 75 minutes, which fits the mobile coverage problem between the district towns. Table 1 also gives the start time of each sensor's longest gap. I do not fill long gaps, because I cannot know what the temperature was during that time. They stay empty and are flagged, so the later models do not assume the milk was fine.

Sensor	Missing %	Longest gap (min)	Longest gap started	Spikes removed
TS-CR-1	0.90	10	2025-04-12 05:45	13
TS-CR-2	0.81	10	2025-06-11 23:00	13
TS-D01	0.87	10	2025-04-12 22:05	17
TS-D02	0.76	10	2025-04-22 09:10	15
TS-D03	0.84	15	2025-08-06 06:35	13
TS-D04	0.80	10	2025-05-30 18:50	14
TS-D05	0.83	10	2025-05-30 20:15	13
TS-U01	0.80	10	2025-09-02 01:55	9
TS-U02	0.75	10	2025-07-30 02:40	11
TS-U03	0.82	10	2025-04-26 17:10	21
TS-U04	0.86	10	2025-04-04 04:00	13
TS-U05	0.86	10	2025-05-23 11:35	12
TS-U06	0.84	10	2025-04-08 14:40	17
TS-U07	0.77	10	2025-05-07 21:30	14
TS-U08	0.78	10	2025-04-06 19:30	15
TS-U09	0.75	10	2025-04-27 02:15	12
TS-U10	0.77	10	2025-05-23 03:20	7
TS-U11	0.80	10	2025-07-14 00:30	8
TS-U12	1.77	2760	2025-08-09 13:00	9
TS-U13	0.73	5	2025-04-01 10:55	8
TS-U14	0.75	10	2025-06-05 03:20	9
TS-U15	0.79	10	2025-06-05 07:20	16
TS-V1	2.95	15	2025-06-09 02:35	17
TS-V2	2.91	15	2025-07-03 20:10	15
TS-V3	4.26	75	2025-04-07 06:30	18

Table 1 – Missing readings, longest gap (length and start time) and removed spikes per sensor.

3. Spikes

I call a reading a spike when it is more than **10 °C** away from the median of the 7 readings around it (35 minutes). A single reading that jumps by 10 to 25 °C and is back to normal 5 minutes later is not physically possible. My first try was 5 °C, but on the vehicle loggers this removed normal temperature changes: V3 had 147 'spikes' while the fixed sensors had 10 to 20. When I raised the threshold, the vehicle counts dropped and levelled off near 10 °C, close to the other sensors (Figure 1). On the fixed sensors every spike I found was between 10.6 and 25 °C, so 10 °C does not cut into normal data. In total 329 spikes were removed (7 to 21 per sensor). Removed readings become gaps. Gaps up to 30 minutes are filled by linear interpolation, longer ones stay empty and are flagged in the cleaned file. One thing to keep in mind: a spike smaller than 10 °C would not be caught.

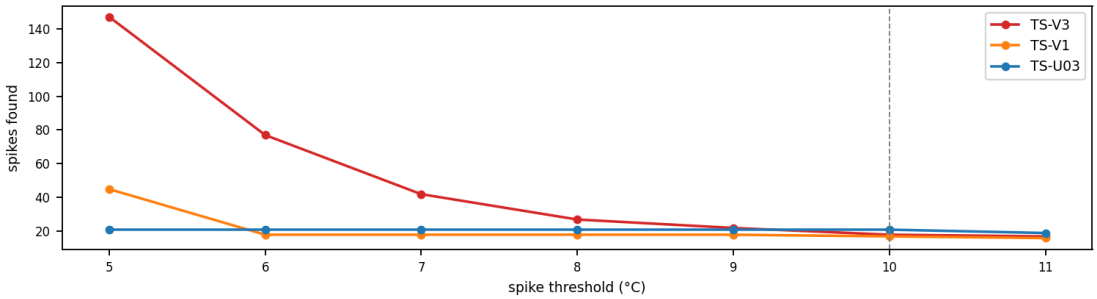


Figure 1 – Spikes found for different thresholds. I used 10 °C.

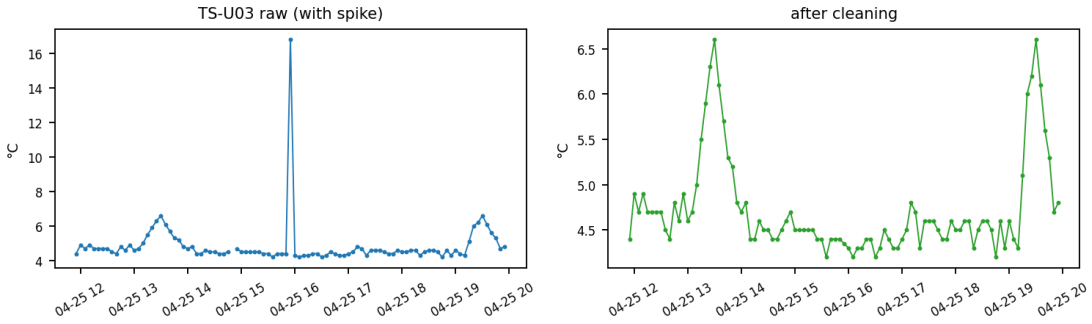


Figure 2 – A spike in TS-U03, raw (left) and after cleaning (right).

4. Sensors that behave abnormally

How I looked for them. For every sensor I took the daily median temperature and subtracted the median of its peer group (the other cabinets, the other vehicles, or the cold-room pair). Heat waves and seasons affect all peers in the same way, so they cancel out, and a fixed calibration offset of about 0.25 °C only moves the line up or down. Then I fitted two simple models for every possible start day, a sudden step and a flat line followed by a slow ramp, and kept the better fit. I flag a sensor when the change is at least 0.6 °C and the model explains at least half of the variation. A gap of 24 hours or more is reported as an outage. For the vehicles I only used readings between the first loading and the last unloading of the day, because a parked van reads outdoor temperature.

Sensor	Problem	Starts	Size	How I know
TS-U07	Slow drift, reads too warm	about 1 Jun 2025	+2.3 °C by 30 Sep (roughly 0.6 °C per month)	The fit explains 99 % of the variation and the other cabinets stay flat
TS-U12	Logger offline	9 Aug 2025, 13:00	46 hours (back on 11 Aug, 10:55)	One gap of 2,760 minutes, then normal data

Table 2 – Abnormal sensors I found.

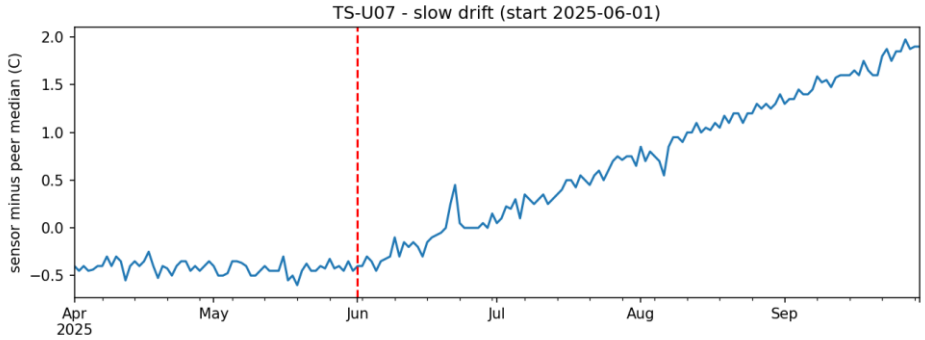


Figure 3 – TS-U07 compared with its peer cabinets. The red line is the estimated start of the drift.

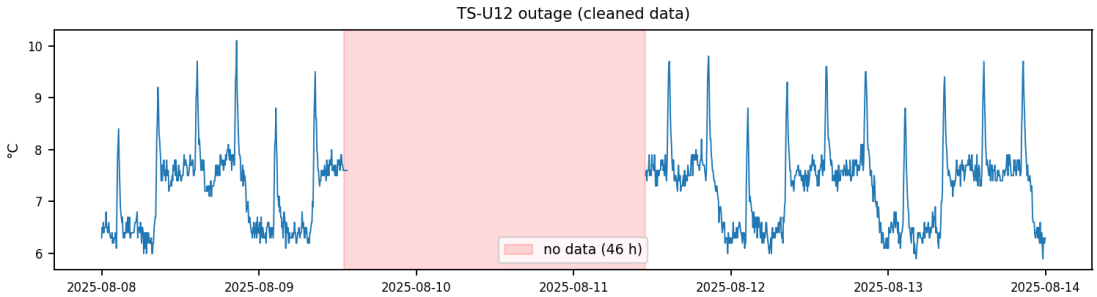


Figure 4 – The 46-hour outage of TS-U12. Its battery level at the end (56 %) is similar to the other sensors (55 to 79 %), so the battery is not the reason.

Looked normal: the other 18 store sensors and both cold-room sensors. The front sensor of the cold room reads about 0.46 °C warmer than the back sensor in every month, which is the door effect, and this difference does not change.

Vehicles, not clear. The on-board temperatures of all three vehicles go up from about 5 to 6 °C in April to 10 to 15 °C in July and August, which is expected with the heat and with loading before the cargo area is cold. Because of this the vehicle data is noisy and I could not detect a clear fault with my method. The only hint is that the difference between V1 and V2 changes by about 1.3 °C around mid-July, but this explains only 26 % of the variation, so I do not call it a fault. It is something I would like to check against the supervisor's fault list.